SQL Data Warehouse & Analytics Project Report

Ridmi Jayawardhana

1. Introduction

In the modern data-driven world, organizations rely heavily on data to support decision-making, improve performance, and gain competitive advantages. However, raw data collected from multiple sources is often unstructured, inconsistent, and difficult to analyze directly. Therefore, structured approaches such as Data Warehousing and Data Analytics are essential.

This project focuses on designing and implementing a complete end-to-end Data Warehouse and Analytics solution using SQL Server. It demonstrates how raw data from multiple systems can be transformed into meaningful insights through a structured pipeline.

The project integrates key concepts such as:

- Data Warehousing
- ETL (Extract, Transform, Load) processes
- Data Modeling
- Exploratory Data Analysis (EDA)
- Advanced Analytical Techniques

By following a systematic approach, this project replicates a real-world data engineering and analytics workflow.

2. Project Objectives

The primary objective of this project is to develop a fully functional data warehouse system and perform analytical operations on it. The specific objectives include:

- To understand and apply Data Warehouse concepts in a practical scenario
- To design a scalable and structured data architecture
- To implement ETL processes for handling raw data
- To clean, transform, and standardize datasets
- To design and implement a dimensional data model using fact and dimension tables
- To perform exploratory and advanced data analysis using SQL
- To generate meaningful business insights from the data
- To prepare structured datasets for reporting and visualization

3. Project Overview

This project is structured into multiple phases, each focusing on a specific aspect of data engineering and analytics. The workflow follows a logical progression from data collection to insight generation.

Project Phases

1. Project Planning and Requirement Analysis
2. Data Architecture Design
3. Project Setup and Initialization
4. Data Ingestion (Bronze Layer)
5. Data Cleaning and Transformation (Silver Layer)
6. Data Modeling (Gold Layer)
7. Exploratory Data Analysis (EDA)
8. Advanced Data Analysis
9. Reporting and Visualization
10. Version Control and Finalization

Each phase builds upon the previous one, ensuring a structured and efficient workflow.

4. Project Planning & Requirement Analysis

This phase serves as the foundation of the entire project. It involves understanding the problem domain, defining objectives, and preparing a roadmap for execution.

Initially, different types of SQL and data-related projects were studied to understand common workflows and best practices. This helped in identifying the most suitable approach for this project.

A detailed study of Data Warehousing (DWH) was conducted to understand how large volumes of structured data can be stored and optimized for analytical queries. Key concepts such as data integration, subject-oriented design, and time-variant data storage were explored.

The ETL (Extract, Transform, Load) process was analyzed to understand how data moves from source systems to the data warehouse. This included:

- Extracting data from raw files
- Transforming data into a usable format
- Loading data into structured tables

The project datasets were then downloaded and examined. This included analyzing:

- Data formats (CSV/Excel)
- Column structures

- Relationships between datasets
- Data quality issues (missing values, duplicates, inconsistencies)

Based on this analysis, the project scope was defined. The system aims to transform raw CRM and ERP datasets into a structured data warehouse and perform analysis to generate insights.

A project plan was created to organize tasks, define timelines, and ensure smooth execution throughout the project lifecycle.

5. Data Architecture Design

The data architecture defines how data is organized, stored, and processed within the system.

For this project, the Medallion Architecture was selected due to its simplicity, scalability, and effectiveness in managing data transformation stages. This architecture divides the system into three layers:

5.1 Bronze Layer (Raw Data Layer)

This layer stores raw data exactly as received from source systems. No transformations are applied at this stage. The purpose is to preserve original data for traceability and auditing.

5.2 Silver Layer (Cleaned Data Layer)

In this layer, data is cleaned, validated, and standardized. Common data quality issues such as:

- Missing values
- Duplicate records
- Inconsistent formats

are handled. The data becomes structured and ready for further processing.

5.3 Gold Layer (Analytical Layer)

This layer contains highly structured, aggregated, and business-ready data. It is optimized for querying and reporting purposes.

Data Flow Design

Data flows sequentially from:
Source Systems → Bronze Layer → Silver Layer → Gold Layer

Each stage applies transformations that improve data quality and usability.

An architecture diagram was created using Draw.io to visually represent:

- Data sources
- Data flow
- Transformation stages
- Storage layers

This diagram serves as a blueprint for the system.

6. Project Setup & Initialization

This phase focuses on preparing the development environment.

A database was created using SQL Server to store and manage data. To maintain proper organization, schemas were defined for each layer:

- Bronze schema
- Silver schema
- Gold schema

Naming conventions were established to ensure consistency across database objects. These conventions included:

- Table naming standards
- Column naming formats
- Schema naming guidelines

Version control was implemented using GitHub. A repository was created to:

- Store SQL scripts
- Track changes
- Maintain project history

The project structure was organized into folders for:

- SQL scripts
- Documentation
- Data files

Initial scripts and configurations were committed to GitHub to establish a baseline.

7. Bronze Layer – Data Ingestion

The Bronze layer is responsible for ingesting raw data into the system.

The first step involved analyzing the source systems, which included CRM and ERP datasets. Understanding these datasets was essential to correctly map them into database tables.

DDL (Data Definition Language) scripts were created to define raw tables. These tables were designed to closely match the structure of the source data.

Data loading was performed using SQL techniques such as:

- Bulk Insert
- Import from flat files

Stored procedures were developed to automate the data loading process. This ensures:

- Consistency
- Reusability
- Efficiency

No transformations were applied in this layer. The goal was to preserve the original data exactly as received.

Data flow from source systems to the Bronze layer was documented for clarity and traceability.

8. Silver Layer – Data Cleaning & Transformation

The Silver layer focuses on improving data quality and preparing it for analysis.

Data from the Bronze layer was explored to identify quality issues such as:

- Null values
- Duplicate entries
- Incorrect formats

Data cleaning techniques were applied, including:

- Removing duplicates
- Handling missing values
- Standardizing formats
- Correcting inconsistencies

Cleaned tables were created to store processed data.

SQL queries and stored procedures were used to implement transformation logic. This ensured that the process is repeatable and automated.

The output of this layer is structured, consistent, and reliable data.

9. Gold Layer – Data Modeling

The Gold layer is designed for analytical purposes.

A Star Schema was implemented to organize data into:

- Fact tables (measurable data)
- Dimension tables (descriptive data)

Dimension Tables

- Customers
- Products

Fact Table

- Sales

Relationships between fact and dimension tables were defined using keys.

This structure allows:

- Fast query performance
- Easy aggregation
- Simplified reporting

A data catalog was created to document:

- Table structures
- Column descriptions
- Relationships

10. Conclusion

This project successfully demonstrates the complete lifecycle of a data warehouse. The project highlights practical skills in:

- SQL development
- Data transformation
- Data modeling
- Analytical thinking

It provides a strong foundation for real-world data engineering and analytics projects.